CUSTOMER CHURN ANALYSIS FOR A TELECOMMUNICATIONS COMPANY

Group 7

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Introduction

This report presents the results of the Customer Churn Analysis project for a telecommunications company, drawing on the team's data preparation, customer segmentation (K-Means clustering) and churn prediction (artificial neural network) work. It has four parts: the summary of key findings, the factors contributing to churn and retention, recommendations for targeted retention strategies, and limitations with proposed solutions.

Key findings at a glance

- **New customers on high monthly charges are the clearest risk.** The New, High Spend segment is 31.6% of training customers and churns at 48.9%, compared with 25.4%, 15.6% and 5.0% for the other three segments.
- **Tenure is the strongest pattern.** Churn falls from 49.0% in the first 12 months to 9.4% beyond four years, and the ANN relies far more on tenure, charges and derived spending (mean AP decrease 0.310) than on any other group (0.018 or less).
- **The ANN is useful for prioritising contacts, not for overall accuracy.** On 1,349 test customers it caught 288 of 358 churners (recall 80.45%) with 342 false alerts (precision 45.71%), and its ROC AUC is 0.8127. Its 69.46% accuracy is below the 73.46% always-no-churn baseline.
- **Contract type alone does not separate churn** (25.2% to 27.7%) in the development data, while churn falls steadily with tenure.
- **No retention benefit has been demonstrated.** The recommended next step is a controlled pilot with a control group, with a fairness review before any age-related targeting.

Part 1: Summary of Key Findings

Data and approach

The analysis uses Dataset_ATS_v2.csv, which holds 7,043 customer records and 10 fields: gender, SeniorCitizen, Dependents, tenure, PhoneService, MultipleLines, InternetService, Contract, MonthlyCharges and Churn. There were no missing values. A total of 302 exact duplicate rows were removed (541 rows belonged to duplicate groups, of which 302 were surplus copies), leaving 6,741 records. These were split into 5,392 training and 1,349 test records, and numeric features were standardised using a scaler fitted on training data only.

Two complementary analyses were run on this data. Customer segmentation (K-Means clustering) describes who the customers are. Churn prediction (an artificial neural network, ANN) estimates which customers are most likely to leave. For the ANN, the 5,392 training records were divided into 4,313 for fitting and 1,079 for validation. The configuration and decision threshold were selected using validation data only, and the 1,349 test records were used for final evaluation.

Customer segmentation

K-Means was applied to standardised tenure and monthly charges. The number of clusters was set to four, supported by two methods. Under the elbow method, the improvement in within-cluster variance

fell from 37.0% at k=4 to 17.9% at k=5. The silhouette score peaked at k=4 (0.4593). The model was fitted on the 5,392 training customers. Churn was not used to form the clusters; churn rates were calculated afterwards.

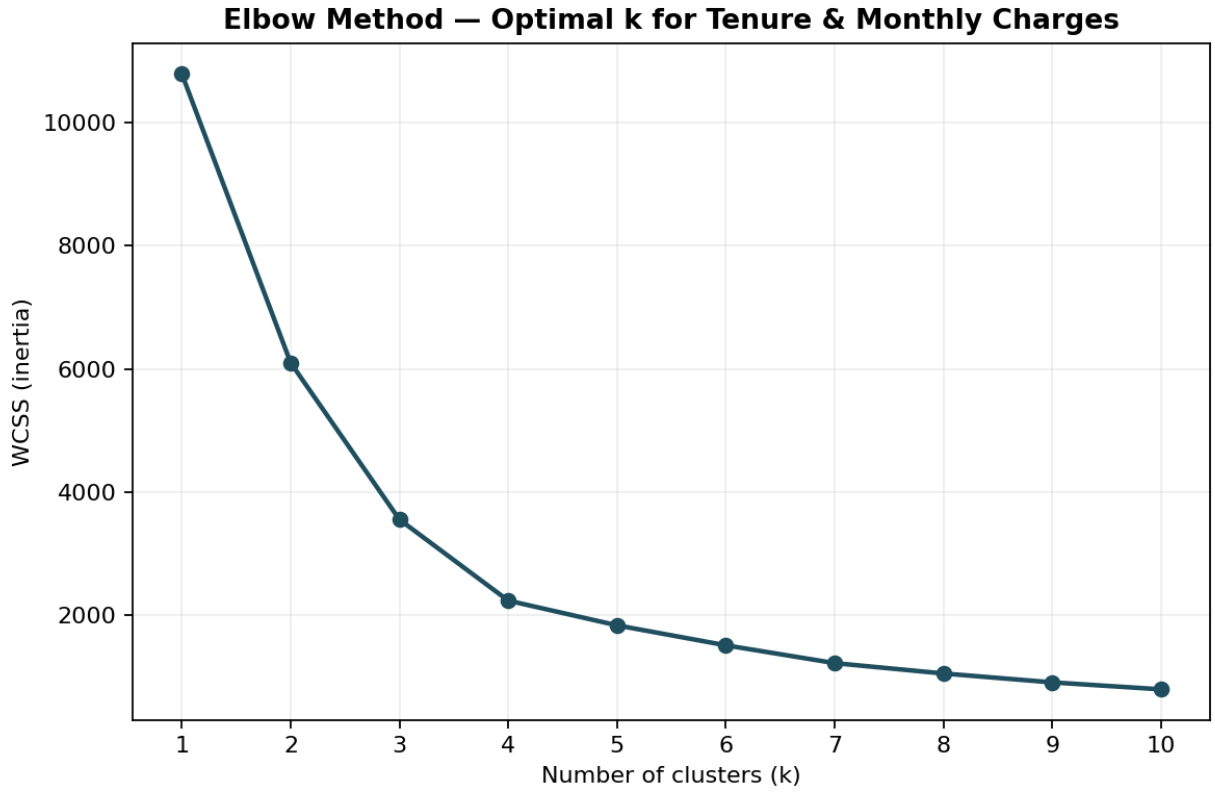


Figure 1. Elbow method: within-cluster sum of squares (WCSS) by number of clusters, k = 1 to 10.

Segment	Profile	Training customers	Training churn rate
Loyal, Low Spend	High tenure, low charges	918	5.0%
Loyal, High Spend	High tenure, high charges	1,521	15.6%
New, High Spend (At-Risk)	Low tenure, high charges	1,705	48.9%
New, Low Spend	Low tenure, low charges	1,248	25.4%

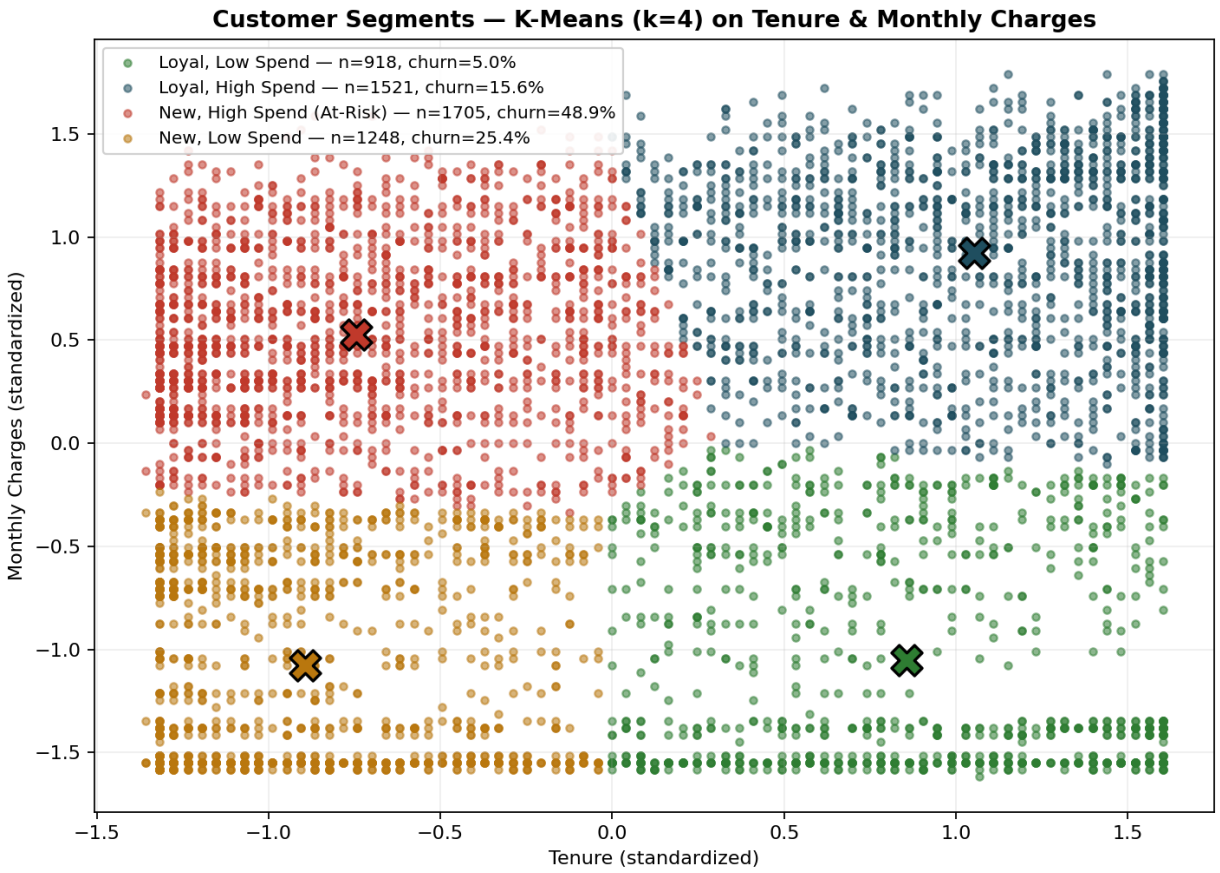


Figure 2. Labelled customer segments: tenure against monthly charges (both standardised), with cluster centroids marked.

These churn rates are descriptive. They were calculated on the same 5,392 training customers used to fit the clusters, and they show association, not cause.

The New, High Spend segment is the clear priority. It churns at 48.9%, compared with 25.4% for New, Low Spend, 15.6% for Loyal, High Spend and 5.0% for Loyal, Low Spend. It is also the largest segment, with 1,705 of the 5,392 training customers (31.6%).

Churn prediction

The selected model is configuration weighted_regularised_32_16, a feed-forward ANN that gives each customer a churn score between zero and one. Customers scoring 0.45 or higher are flagged as likely to churn.

Component	Setting
Input layer	16 encoded numerical features, including the engineered spending, service-count and tenure-group features
Hidden layers	32 and 16 units, ReLU activation
Output	One sigmoid neuron (churn score, not independently calibrated)

Component	Setting
Regularisation	Dropout of 0.2 after each hidden layer; L2 coefficient of 0.001 on hidden-layer weights
Class weighting	Balanced class weights
Training	Adam optimiser (learning rate 0.001), binary cross-entropy loss, batch size 32, up to 80 epochs
Early stopping	On validation loss, patience 10, best weights restored (epoch 41 retained, 51 epochs run)
Trainable parameters	1,089
Data split	4,313 fitting, 1,079 validation, 1,349 test; scaler fitted on the fitting subset only

Three configurations (baseline, regularised, and weighted regularised) were each trained with seeds 42, 43 and 44. Selection maximised mean validation F1 over thresholds from 0.20 to 0.70 in steps of 0.05. The selected pair scored a mean validation F1 of 0.5973 (seed SD 0.0140), and the delivered model uses seed 42.

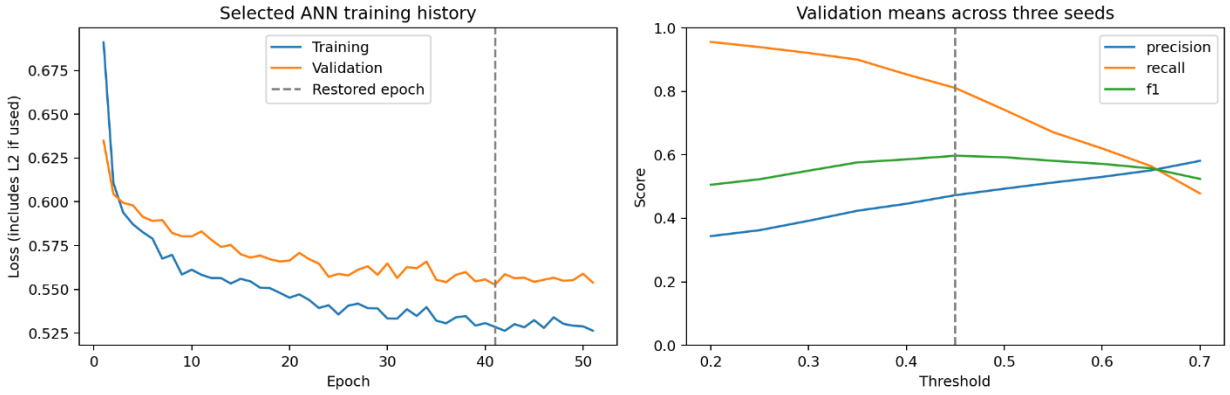


Figure 3. Training and validation loss for the selected ANN with the restored epoch marked (left), and validation precision, recall and F1 by threshold, averaged across three seeds (right).

Metric	ANN (test)	95% bootstrap interval	Always predict no churn
Accuracy	69.46%	66.94% to 71.68%	73.46%
Precision	45.71%	42.05% to 49.53%	0.00%
Recall	80.45%	76.15% to 84.49%	0.00%
F1-score	58.30%	54.66% to 61.88%	0.00%

	Predicted churn	Predicted no churn
Actual churn (358)	288 caught	70 missed
Actual no churn (991)	342 false alerts	649 correct

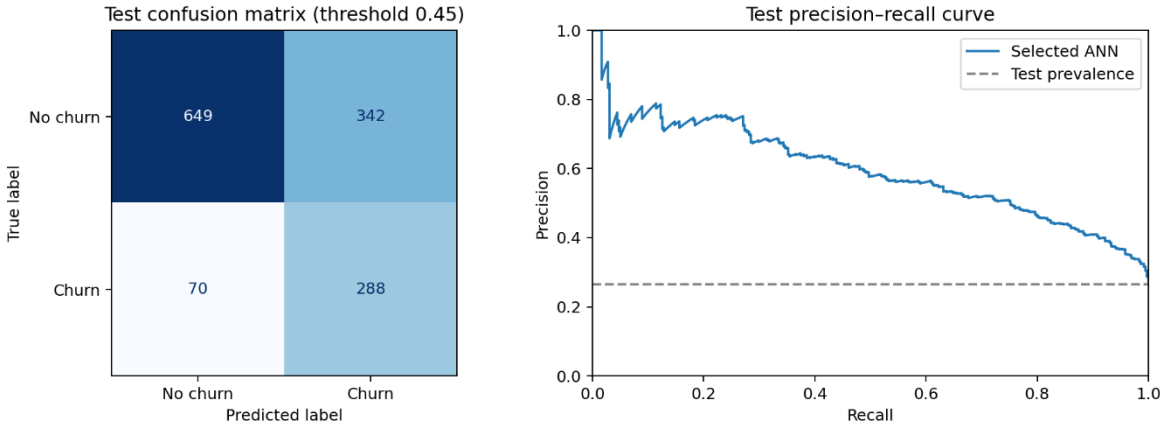


Figure 4. Test confusion matrix at threshold 0.45 (left) and test precision-recall curve against test prevalence (right).

How to read these results:

- The model caught 288 of the 358 customers who churned (80.45%) and missed 70.
- It flagged 630 of 1,349 test customers (46.7%). Of those, 45.7% actually churned, compared with a 26.5% churn rate across all test customers.
- Accuracy needs care. The 69.46% result is lower than the 73.46% accuracy of always predicting no churn, but that baseline identifies no churners at all. The model gives up some overall accuracy to catch most churners, so recall and precision, not accuracy, are the meaningful measures here. Two threshold-free measures show that the scores rank customers well above chance: the test ROC AUC is 0.8127 and the average precision is 0.5957, against a 26.5% base rate.
- Robustness: The 95% bootstrap intervals are in the table above. The whole accuracy interval (66.94% to 71.68%) sits below the 73.46% baseline, while the recall interval (76.15% to 84.49%) stays high. Bootstrap intervals describe test-sample uncertainty for this fixed model only, and variation across seeds should not be read as cross-validation.

How the two analyses fit together

Segmentation and prediction were developed independently, yet both point to tenure and spending. Churn is highest for new customers on high charges, and the group of tenure, charges and derived spending is the group the model relies on most, by a wide margin. These are associations, not proof of cause.

Part 2: Factors Contributing to Churn and Retention

Everything in this part describes association. It shows who churns more often and what the model relies on, not why customers leave.

Evidence from the segmentation

- Tenure: within each spend level, newer customers churn far more. At low spend, churn is 25.4% for new customers against 5.0% for loyal customers. At high spend, it is 48.9% against 15.6%.
- Monthly charges: within each tenure level, higher charges go with higher churn. Among new customers, churn is 48.9% at high charges against 25.4% at low charges. Among loyal customers, it is 15.6% against 5.0%.
- Combined: churn is lowest for long-tenure, low-charge customers and highest for new customers on high charges. High and low are relative to the training average.

Evidence from the ANN

Grouped importance was measured by permutation on the validation set. All columns in a feature group were shuffled together, and the mean decrease in average precision (AP) was recorded over 10 shuffles, shown with its standard deviation (SD) across those shuffles. The tenure, charges and derived spending group holds tenure, MonthlyCharges, EstimatedTotalSpend, NumActiveServices, AvgMonthlyChargePerService and the tenure-group columns. The phone and multiple lines group holds the phone service and multiple lines columns.

Feature group	Mean AP decrease	SD
Tenure, charges and derived spending	0.3097	0.0130
Phone and multiple lines	0.0179	0.0067
Senior citizen	0.0076	0.0085
Contract	0.0042	0.0027
Internet service	0.0013	0.0026
Dependents	0.0010	0.0079
Gender	-0.0002	0.0054

The model relies overwhelmingly on tenure, charges and derived spending: its decrease (0.3097) is far larger than that of the next group, phone and multiple lines (0.0179). The remaining decreases are small. For senior citizen, internet service and dependents the SD is larger than the mean decrease, so their order should not be over-read, and gender shows no positive decrease. Correlated engineered features were grouped, so their individual effects cannot be separated. These results show what the model relies on, not proof of causal drivers or of the direction of influence.

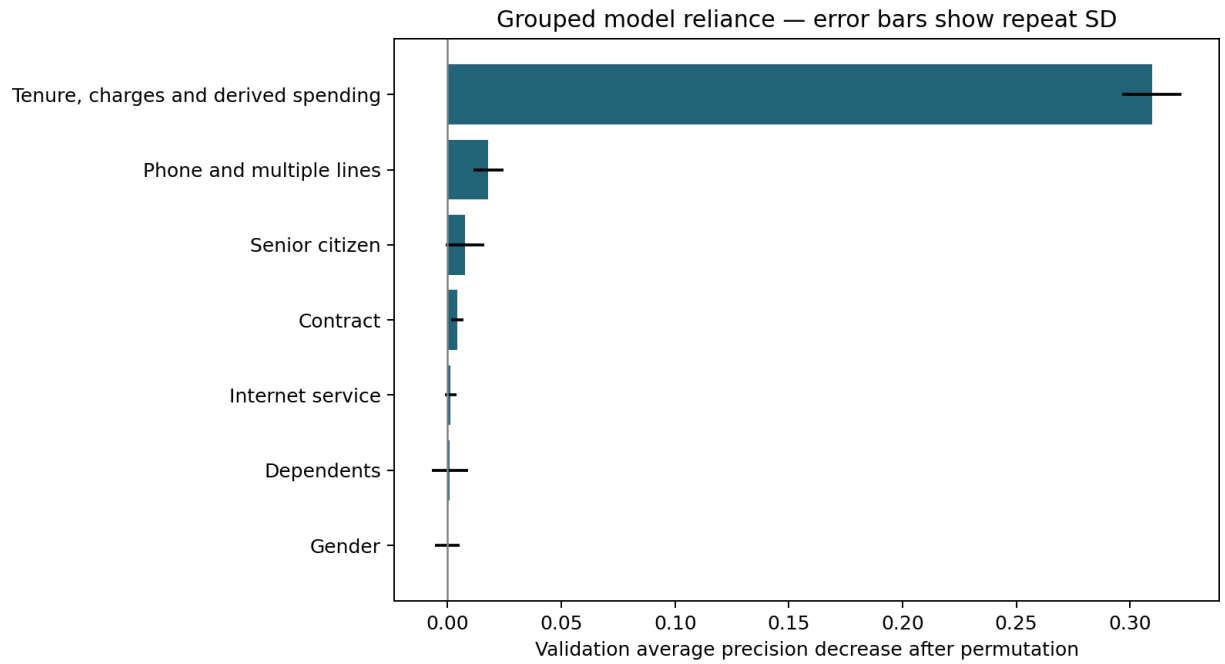


Figure 5. Grouped model reliance: decrease in validation average precision after permutation, with error bars showing the repeat SD.

Contract type and tenure bands

Observed churn rates among the 5,392 development customers (training and validation records), taken from development_customer_profiles.csv:

Factor	Category	Customers	Observed churn rate
Tenure band	0 to 12 months	1,579	49.0%
Tenure band	13 to 24 months	817	28.6%
Tenure band	25 to 48 months	1,253	20.9%
Tenure band	49+ months	1,743	9.4%
Contract group	One year	1,153	25.2%
Contract group	Month-to-month	2,877	27.7%
Contract group	Two year	1,362	25.4%

Churn falls steadily as tenure rises, from 49.0% in the first year to 9.4% beyond four years. By contrast, the three contract groups sit within about 2.5 percentage points of each other (25.2% to 27.7%), so contract group on its own does not separate churn in these data. That matches its small grouped importance (0.0042).

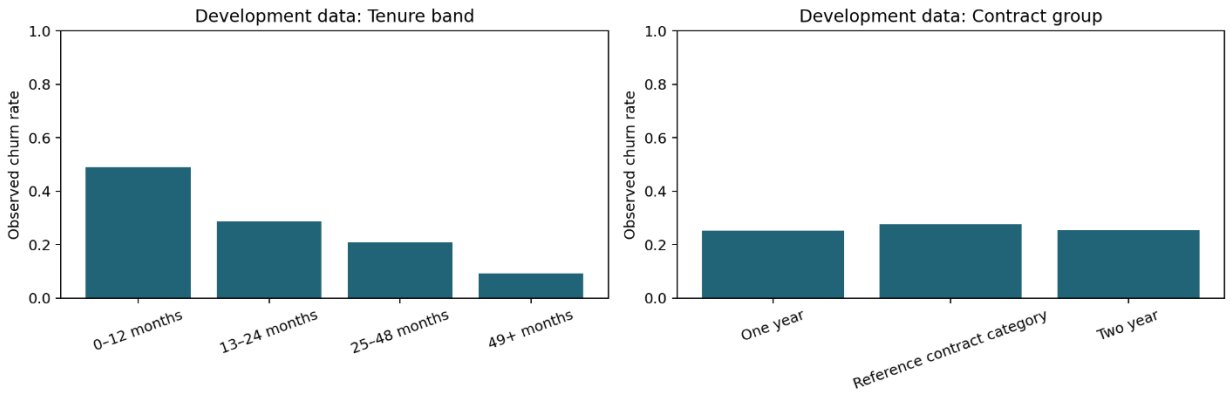


Figure 6. Observed churn rate by tenure band (left) and contract group (right) in the development data. In the contract chart, the bar labelled Reference contract category is the Month-to-month group.

Factors associated with retention

- Long tenure with lower monthly charges is the strongest retention profile: 5.0% churn among training customers.
- Long tenure still helps at higher charges. Loyal, High Spend customers churn at 15.6%, against 48.9% for new customers at similar charge levels.
- Service quality, price perception, competitor offers and customer experience are not among the 10 fields, so the data cannot say why retained customers stay.

Part 3: Recommendations for Targeted Retention Strategies

These recommendations follow from observed patterns and model scores. No retention uplift or financial benefit has been demonstrated, so each should be tested before it is scaled.

1. **Start with a small, controlled retention pilot.** Use the churn score to rank customers and contact those with the highest scores, within a defined contact budget. Hold out a comparable control group that is not contacted, and compare churn between the two groups to measure whether the intervention works. Judge the pilot on the churn difference between the groups, the number of contacts needed per retained customer, and the cost of the offers, and agree a review point before it starts.
2. **Set the decision threshold with the business.** At the current threshold of 0.45 the model flags 630 of 1,349 test customers, including 342 false alerts, and misses 70 of 358 churners. Before operational use, review the workload that false alerts create and the cost of missed churners, and choose a threshold that fits the contact budget. The table after this list shows how the trade-off looks on validation data.
3. **Treat New, High Spend customers as the first group to test.** This segment holds about 32% of training customers and has the highest churn rate of the four segments (48.9%). Churn is also concentrated early in the customer lifetime: 49.0% of customers in their first 12 months churned, against 9.4% beyond four years. Early-tenure ideas worth testing include a proactive plan and bill review in the first months, or a welcome check-in. These are hypotheses, not proven fixes, because the data do not show why these customers leave.

4. **Offer incentives only where the evidence supports them.** Feature importance does not justify a specific discount or show why customers left. Offers or service reviews should be considered only where the observed profiles and business context support them. Loyal, Low Spend customers churn at about 5%, which leaves little room for improvement, so they are a lower priority for retention spend.
5. **Review fairness before deployment.** Gender and age-related inputs, such as senior citizen status, should not automatically determine who receives an offer. Review suitability, and model performance across relevant groups, before deployment.
6. **Use the score and the segments together, then monitor.** The score identifies who to contact; the segments suggest what kind of conversation to have. After launch, track model performance, data drift and subgroup results.

Threshold trade-off on validation data

The figures below are means over three seeds for the selected configuration, from validation data rather than test results. Validation F1 is nearly flat from 0.40 to 0.55 (0.5812 to 0.5973), so the threshold is best set by contact capacity and the cost of errors rather than by F1 alone.

Threshold	Accuracy	Precision	Recall	F1
0.35	64.78%	42.38%	90.01%	57.62%
0.40	67.87%	44.59%	85.37%	58.57%
0.45 (selected)	70.90%	47.29%	81.07%	59.73%
0.50	72.85%	49.32%	74.22%	59.25%
0.55	74.27%	51.28%	67.13%	58.12%
0.60	75.29%	53.05%	62.02%	57.16%

Part 4: Limitations and Proposed Solutions

Data and modelling limitations

Limitation	Why it matters	Proposed solution	Feasible within this project?
One reduced historical dataset (10 fields; no billing history, payment method, service	Limits which churn drivers the model can see	Collect richer, time-stamped features	Future work, needs new data

Limitation	Why it matters	Proposed solution	Feasible within this project?
quality or contact history)			
Deduplication without customer IDs	The 302 removed rows were exact duplicates on all 10 fields, but without IDs true duplicates cannot be told apart from different customers with identical attributes	Confirm the duplicate policy with the Data Engineer and obtain customer IDs	Partly: the policy can be confirmed now, IDs need the data owner
Single fixed train, validation and test split	Results may depend on how the data happened to be split, and variation across seeds is not cross-validation	Repeat with cross-validation or repeated splits	Doable now
Possible validation selection optimism	The configuration and threshold were chosen because they scored best on validation data, so validation scores may flatter them	Test results are reported for the final model; confirm on genuinely new data	Future work, needs new data
Correlated features	Engineered features overlap with tenure and charges, so individual effects cannot be separated	Keep grouped reporting and test with feature ablation if needed	Doable now
Uncalibrated sigmoid scores	Scores rank customers but are not reliable probabilities	Assess calibration before using scores in cost calculations	Doable now, on validation data
No temporal or external validation, and no causal evaluation	Future performance is unproven, and the model cannot show that an intervention reduces churn	Evaluate on future data and run a controlled pilot with a control group	Future work, needs future data and a pilot
Segmentation uses only two features	K-Means assumes roughly round, similar-sized clusters and is sensitive to outliers; segments were fitted on the 5,392 training customers and churn rates are descriptive	Test more features and alternative algorithms such as hierarchical clustering, and check stability across seeds and samples	Doable now

Limitation	Why it matters	Proposed solution	Feasible within this project?
Engineered features are proxies	In the feature-engineering script, EstimatedTotalSpend (tenure x MonthlyCharges) stands in for TotalCharges, which the dataset lacks	Use actual billing totals if obtainable	Future work, needs new data
Final model not refit on all training rows	The model was trained on 4,313 rows. The 1,079 validation records were used only to choose the configuration and threshold, so they did not help train the delivered model	Refit with the selected settings on all 5,392 training customers, then re-evaluate	Doable now, but the results would need re-evaluating

Project challenges and how they were handled

Challenge	How it was handled
Confluence was not enabled on the team's Atlassian site	Jira, GitHub and Word documents were used for tracking and documentation
A compressed Stage 3 timeline, with the report and video depending on the ANN results	Bi-weekly checkpoints, early sharing of draft metrics, and drafting the report in parallel
K-Means cluster numbers are arbitrary and can change between runs or environments	Segment labels are derived from centroid values and churn rate rather than a fixed lookup by cluster number
Duplicate counts could be read two ways (302 removable rows, 541 rows in duplicate groups)	Both definitions were documented so the figures are cited consistently
Early ANN draft results were later superseded	The final run corrected the scaling used for internal validation, compared three configurations across three seeds, chose the threshold on validation data, and used the test set for one final evaluation. The earlier draft figures (78.04% accuracy, 44.25% recall) are superseded by the final results

The proposed solutions above are future actions, not completed checks.

Conclusion

This project combined customer segmentation and churn prediction to understand who leaves and how retention effort could be focused. K-Means clustering on tenure and monthly charges produced four customer segments, and new customers on high monthly charges had the highest churn rate (48.9%). The ANN identified 288 of the 358 churners in the test set (recall 80.45%), at the cost of 342 false alerts

(precision 45.71%). Its accuracy of 69.46% is below the 73.46% always-no-churn baseline, so the model is best used to prioritise which customers to contact, not to predict every customer correctly. Both analyses point to tenure and spending as the factors most closely associated with churn.

These results describe associations, not causes, and come from a single reduced dataset with a fixed split, so they are a starting point rather than a final answer. The recommended next step is a small, controlled retention pilot with a control group, a threshold set to fit the contact budget, and a fairness review before any offers are targeted. Addressing the remaining limitations will need richer, time-stamped data and evaluation on future customers, alongside the checks that can be done now, such as cross-validation and calibration.